Supplementary file 3: List of studies excluded after full text review with reasons

List of studies excluded after full text review with reason for exclusion (n=72)

Summary

24 – Not a systems approach
21 – Abstracts, posters, protocols, reviews and duplicate
10 – Problem not framed in a systems context
5 – No comparator
4 – No quantitative data analysed
2 – Not been explicit about a systems approach
2 – Simulation results not applied in real-life
1 – Framed in a systems context but not evident in paper
1 – Not enough details on intervention
1 – Framework developed, not primary research
1 – About education in systems thinking

	Reference of excluded full text	Reason for exclusion
1	Dusek, J. A. *et al.* (2016) 'Patients Receiving Integrative Medicine Effectiveness Registry (PRIMIER) of the BraveNet practice-based research network: study protocol', *BMC Complementary and Alternative Medicine*, 16(1), p. 53. doi: 10.1186/s12906-016-1025-0.	Study protocol
2	Minkman, M., Ahaus, K. and Huijsman, R. (2007) 'Performance improvement based on integrated quality management models: what evidence do we have? A systematic literature review', *International Journal for Quality in Health Care*, 19(2), pp. 90–104. doi: 10.1093/intqhc/mzl071.	Literature review
3	Dhruva, A. *et al.* (2014) 'A Prospective Clinical Study of a Whole Systems Ayurvedic Intervention for Breast Cancer Survivorship', *The Journal of Alternative and Complementary Medicine*, 20(5), pp. A72–A72. doi: 10.1089/acm.2014.5189.abstract.	Abstract
4	Woods, A. (2008) 'Using lean/six sigma methodology to decrease error rate and cost of quality', *Transfusion,* Vol. 58 (supplement 2)	Poster
5	Dunbar, J. A.; O'Reilly, D. A. R.; Versace, V.; Sophy, S.; Janus, E. D (2017) Preventing progression to type 2 diabetes in women who have had gestational diabetes: Back to the drawing board?, European Association for the Study of Disease virtual meeting.	Poster
6	Boustani, M. A. (2017) Implementing the collaborative dementia care model in the real world.	Poster
7	Chandiramani, M. J.; … (2019) A multidisciplinary, multi-faceted approach to redesigning care pathways in the maternity assessment unit.	Abstract
8	P. W. Mirhosseini, C.;Hayes-Bautista, T. (2018) Depression screening: A "systems thinking" approach to address health disparities in ob/gyn practice	Abstract
9	S. J. C. Naidu, P.;Rosenthal, M.;Naik, S.;Patel, D.;Sawmynaden, V.;Cummings, S.;Jemmott, A.;Basi, M.;Hacker, K. (2019) An example of strategic collaborative working across a North Central London borough, over a 3-year period, to improve the care for people with diabetes and serious mental illness	Poster abstract
10	S. Y. Bakhai (2018) Implementation of integrated transition of care management in an academic, hospital-based safety-net primary care clinic	Presentation
11	Sherr, K. *et al.* (2014) 'Systems analysis and improvement to optimize pMTCT (SAIA): a cluster randomized trial', *Implementation science : IS*. England, 9, p. 55. Available at: http://ovidsp.ovid.com/ovidweb.cgi?T=JS&PAGE=reference&D=medl&NEWS=N&AN=24885976.	Protocol
12	Schnurr, P. P. *et al.* (2013) 'RESPECT-PTSD: re-engineering systems for the primary care treatment of PTSD, a randomized controlled trial', *Journal of General Internal Medicine*. United States, 28(1), pp. 32–40. Available at: http://ovidsp.ovid.com/ovidweb.cgi?T=JS&PAGE=reference&D=medl&NEWS=N&AN=22865017.	Not been explicit about a systems approach
13	Dietrich, A. J. *et al.* (2004) 'Re-engineering systems for the treatment of depression in primary care: cluster randomised controlled trial', *BMJ (Online)*.	Not been explicit about a systems approach
14	Muder, R. R. *et al.* (2008) 'Implementation of an industrial systems-engineering approach to reduce the incidence of methicillin-resistant Staphylococcus aureus infection', *Infection control and hospital epidemiology*. United States,	Framed in a systems context but not evident in paper

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	29(8), pp. 702–708. Available at: http://ovidsp.ovid.com/ovidweb.cgi?T=JS&PAGE=reference&D=med6&NEWS=N&AN=18624651.	
15	Adesina, A. A. *et al.* (2017) 'Assessing the Value of System Theoretic Process Analysis in a Pharmacovigilance Process: An Example Using Signal Management', *Pharmaceutical Medicine*. Springer International Publishing, 31(4), pp. 267–278. doi: 10.1007/s40290-017-0195-5.	No comparator
16	Alimohammadzadeh, K. *et al.* (2017) 'Assessing common medical errors in a Children's hospital NICU using failure mode and effects analysis (FMEA)', *Trauma Monthly*, 22(5), pp. 1–6. doi: 10.5812/traumamon.15845.	No comparator
17	Arrington-Sanders, R. *et al.* (2018) 'A system-level approach to improve HIV screening in an urban pediatric primary care setting', *Pediatrics*, 142(5). doi: 10.1542/peds.2018-0506.	Not a systems approach
18	Bolton, K. A. *et al.* (2017) 'The outcomes of health-promoting communities: Being active eating well initiative- A community-based obesity prevention intervention in Victoria, Australia', *International Journal of Obesity*. Nature Publishing Group, 41(7), pp. 1080–1090. doi: 10.1038/ijo.2017.73.	Not a systems approach
19	Carrougher, G. J. *et al.* (2017) 'An Intervention Bundle to Facilitate Return to Work for Burn-Injured Workers: Report from a Burn Model System Investigation', *Journal of Burn Care and Research*, 38(1), pp. e70–e78. doi: 10.1097/BCR.0000000000000410.	Not a systems approach
20	Hilton, L. G. *et al.* (2019) 'Evaluation of an Integrative Post-Traumatic Stress Disorder Treatment Program', *Journal of Alternative and Complementary Medicine*, 25(S1), pp. S147–S152. doi: 10.1089/acm.2018.0424.	Not a systems approach
21	Hung, D. Y. *et al.* (2017) 'Scaling lean in primary care: Impacts on system performance', *American Journal of Managed Care*, 23(3), pp. 161–168.	Not enough details on intervention
22	Hussein, N. A. *et al.* (2017) 'Mitigating overcrowding in emergency departments using Six Sigma and simulation: A case study in Egypt', *Operations Research for Health Care*. Elsevier Ltd, 15, pp. 1–12. doi: 10.1016/j.orhc.2017.06.003.	Simulation results not applied in real-life
23	Kazemian, P. *et al.* (2017) 'Coordinating clinic and surgery appointments to meet access service levels for elective surgery', *Journal of Biomedical Informatics*. Elsevier Inc., 66, pp. 105–115. doi: 10.1016/j.jbi.2016.11.007.	Simulation results not applied in real-life
24	Lukes, T., Schjodt, K. and Struwe, L. (2019) 'Implementation of a nursing based order set: Improved antibiotic administration times for pediatric ED patients with therapy-induced neutropenia and fever', *Journal of Pediatric Nursing*. Elsevier Inc., 46, pp. 78–82. doi: 10.1016/j.pedn.2019.02.028.	Problem not framed in systems context
25	Martin, C. M. *et al.* (2019) 'Anticipatory care in potentially preventable hospitalizations: Making data sense of complex health journeys', *Frontiers in Public Health*, 6(JAN). doi: 10.3389/fpubh.2018.00376.	No quantitative data to analyse
26	Mutale, W. *et al.* (2017) 'Application of systems thinking: 12-month postintervention evaluation of a complex health system intervention in Zambia: the case of the BHOMA', *Journal of Evaluation in Clinical Practice*, 23(2), pp. 439–452. doi: 10.1111/jep.12354.	No quantitative data to analyse
27	Myers, M. K. *et al.* (2018) 'Using knowledge translation for quality improvement: An interprofessional education intervention to improve thromboprophylaxis among medical inpatients', *Journal of Multidisciplinary Healthcare*, 11, pp. 467–472. doi: 10.2147/JMDH.S171745.	Problem not framed in systems context though they say they performed a systems analysis
28	Redwood, R. *et al.* (2018) 'Reducing unnecessary culturing: A systems approach to evaluating urine culture ordering and collection practices among nurses in two acute care settings', *Antimicrobial Resistance and Infection Control*. Antimicrobial Resistance & Infection Control, 7(1), pp. 1–8. doi: 10.1186/s13756-017-0278-9.	No quantitative data to analyse
29	Steward, D., Glass, T. F. and Ferrand, Y. B. (2017) 'Simulation-Based Design of ED Operations with Care Streams to Optimize Care Delivery and Reduce Length of Stay in the Emergency Department', *Journal of Medical Systems*. Journal of Medical Systems, 41(10). doi: 10.1007/s10916-017-0804-6.	No baseline data
30	Adaba, G. B. and Kebebew, Y. (2018) 'Improving a health information system for real-time data entries: An action research project using socio-technical systems theory', *Informatics for Health and Social Care*. Taylor & Francis, 43(2), pp. 159–171. doi: 10.1080/17538157.2017.1290638.	No comparator
31	Akhter, L. S. *et al.* (2017) 'Improving Asthma Control through Asthma Action Plans: A Quality Improvement Project at a Midwest Community Clinic', *Journal of Community Health Nursing*. Taylor & Francis, 34(3), pp. 136–146. doi: 10.1080/07370016.2017.1340764.	Not a systems approach
32	Bal, A., Ceylan, C. and Taçoğlu, C. (2017) 'Using value stream mapping and discrete event simulation to improve efficiency of emergency departments', *International Journal of Healthcare Management*, 10(3), pp. 196–206. doi: 10.1080/20479700.2017.1304323.	No comparator in practice, just assume/simulate the future state
33	Losby, J. L. *et al.* (2017) 'Safer and more appropriate opioid prescribing: a large healthcare system's comprehensive approach', *Journal of Evaluation in Clinical Practice*, 23(6), pp. 1173–1179. doi: 10.1111/jep.12756.	Not a systems approach

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34	Verbano, C., Crema, M. and Nicosia, F. (2017) 'Visual management system to improve care planning and controlling: the case of intensive care unit', *Production Planning and Control*. Taylor & Francis, 28(15), pp. 1212–1222. doi: 10.1080/09537287.2017.1358830.	Not a systems approach
35	Yusoff, N. S. M. *et al.* (2018) 'Discrete event simulation and data envelopment analysis models for selecting the best resource allocation alternative at an emergency department's green zone', *Sains Malaysiana*, 47(11), pp. 2917–2925. doi: 10.17576/jsm-2018-4711-35.	Not a systems approach, no comparator
36	Ammenwerth, E. *et al.* (2002) 'Systems analysis in healthcare: framework and example', *Methods of information in medicine*, 41, pp. 134–40.	Framework development not primary research
37	Boden, D. G. *et al.* (2016) 'Lowering levels of bed occupancy is associated with decreased inhospital mortality and improved performance on the 4-hour target in a UK District General Hospital', *Emergency Medicine Journal*, 33(2), pp. 85–90. doi: 10.1136/emermed-2014-204479.	Not framed in a systems context
38	Clark, C. *et al.* (2001) 'A Systematic Approach to Risk Managed Care Environment Improves', *Diabetes care*, 24(6), pp. 1079–1086. Available at: http://care.diabetesjournals.org/content/24/6/1079.full.pdf+html.	Not a systems approach
39	Gaupp, R., Körner, M. and Fabry, G. (2016) 'Effects of a case-based interactive e-learning course on knowledge and attitudes about patient safety: A quasi-experimental study with third-year medical students', *BMC Medical Education*. BMC Medical Education, 16(1), pp. 1–8. doi: 10.1186/s12909-016-0691-4.	About education in systems thinking
40	Gunn, J. *et al.* (2006) 'A systematic review of complex system interventions designed to increase recovery from depression in primary care', *BMC Health Services Research*, 6, pp. 1–11. doi: 10.1186/1472-6963-6-88.	Systematic review
41	Horbar, J. D. *et al.* (2004) 'Collaborative quality improvement to promote evidence based surfactant for preterm infants: A cluster randomised trial', *British Medical Journal*, 329(7473), pp. 1004–1007.	Not framed in a systems context
42	Press, A. I. N. (2005) 'A multifaceted collaborative quality improvement intervention significantly improves delivery of surfactant therapy for preterm infants', *Evidence-Based Healthcare and Public Health*, 9(3), pp. 219–220. doi: 10.1016/j.ehbc.2005.03.014.	Duplicate – same as Horbar et al (2004)
43	Jeon, Y. H. *et al.* (2012) 'Staff outcomes from the Caring for Aged Dementia Care REsident Study (CADRES): A cluster randomised trial', *International Journal of Nursing Studies*. Elsevier Ltd, 49(5), pp. 508–518. doi: 10.1016/j.ijnurstu.2011.10.020.	Not a systems approach
44	Jimmy, L. W. K. *et al.* (2009) 'Reduction in length of hospitalisation for microbial keratitis patients: A prospective study', *International Journal of Health Care Quality Assurance*, 22(7), pp. 701–708. doi: 10.1108/09526860910995038.	Problem not framed in a systems context
45	Kessels-Habraken, M. *et al.* (2010) 'Prospective risk analysis prior to retrospective incident reporting and analysis as a means to enhance incident reporting behaviour: A quasi-experimental field study', *Social Science and Medicine*. Elsevier Ltd, 70(9), pp. 1309–1316. doi: 10.1016/j.socscimed.2010.01.035.	No clear problem framed in a systems context
46	Lin, J. C. and Lee, T. T. (2016) 'Outcomes of medication administration information system for nurses', *Studies in Health Technology and Informatics*, 225(138), pp. 860–861. doi: 10.3233/978-1-61499-658-3-860.	Not a systems approach
47	Macfarlane, F. *et al.* (2013) 'Achieving and sustaining profound institutional change in healthcare: Case study using neo-institutional theory', *Social Science and Medicine*. Elsevier Ltd, 80, pp. 10–18. doi: 10.1016/j.socscimed.2013.01.005.	Not a systems approach, no quantitative results, does not aim to demonstrate effectiveness of SA
48	Mehta, A. D. *et al.* (2010) 'Poster 2: A System Redesign Approach to Improving Timeliness of New Outpatient PM&R Consults: Veterans Affairs Observational Analysis and System Redesign', *Pm&R*. Elsevier Inc., 2(9), pp. S9–S10. doi: 10.1016/j.pmrj.2010.07.033.	Poster
49	Miller, R. S. *et al.* (2010) 'Miller et al-2010-Systems initiatives reduce healthcare-associated infections.pdf', *The Journal of Trauma*, 68(1), pp. 23–31.	Not a systems approach
50	Mills, P. R., Weidmann, A. E. and Stewart, D. (2017) 'Hospital electronic prescribing system implementation impact on discharge information communication and prescribing errors: a before and after study', *European Journal of Clinical Pharmacology*. European Journal of Clinical Pharmacology, 73(10), pp. 1279–1286. doi: 10.1007/s00228-017-2274-7.	Not a systems approach
51	Moody-Thomas, S. *et al.* (2011) 'Awareness and implementation of the 2000 United States public health service tobacco dependence treatment guideline in a public hospital system', *Population Health Management*, 14(2), pp. 79–85. doi: 10.1089/pop.2010.0004.	Not a systems approach
52	Odetola, F. O. *et al.* (2016) 'An innovative framework to improve efficiency of interhospital transfer of children in respiratory failure', *Annals of the American Thoracic Society*, 13(5), pp. 671–677. doi: 10.1513/AnnalsATS.201507-401OC.	Not framed in a systems context
53	Palma, A. *et al.* (2013) 'Applying Systems Dynamics modeling to epidemiological research: an example of PSA screening', *American journal of epidemiology*, 175, pp. 1–145.	Abstract

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54	Procter, S. *et al.* (2013) 'Success and failure in integrated models of nursing for long term conditions: Multiple case studies of whole systems', *International Journal of Nursing Studies*. Elsevier Ltd, 50(5), pp. 632–643. doi: 10.1016/j.ijnurstu.2012.10.007.	Not a systems approach, no comparator
55	Rahman, O. *et al.* (2010) 'Sustained reduction of ventilator associated pneumonia-use of an innovation system process in a tertiary care centre', *Critical care clinics*, 38(12).	Abstract
56	Raupach, T. *et al.* (2014) 'Structured smoking cessation training for health professionals on cardiology wards: A prospective study', *European Journal of Preventive Cardiology*, 21(7), pp. 915–922. doi: 10.1177/2047487312462803.	Not a systems approach
57	Sethi, R. *et al.* (2017) 'A systematic multidisciplinary initiative for reducing the risk of complications in adult scoliosis surgery', *Journal of Neurosurgery: Spine*, 26(6), pp. 744–750. doi: 10.3171/2016.11.SPINE16537.	Not a systems approach
58	Sethi, R. K. *et al.* (2014) 'The Seattle spine team approach to adult deformity surgery: A systems-based approach to perioperative care and subsequent reduction in perioperative complication rates', *Spine Deformity*. Elsevier Inc, 2(2), pp. 95–103. doi: 10.1016/j.jspd.2013.12.002.	New surgical protocol, not a systems approach
59	Singh, R. *et al.* (2012) 'IT-enabled systems engineering approach to monitoring and reducing ADEs', *American Journal of Managed Care*, 18(3), pp. 169–175.	Not framed in a systems context
60	Sobolev, B. G., Sanchez, V. and Vasilakis, C. (2011) 'Systematic review of the use of computer simulation modeling of patient flow in surgical care', *Journal of Medical Systems*, 35(1), pp. 1–16. doi: 10.1007/s10916-009-9336-z.	Systematic review
61	Solberg, L. I. *et al.* (1997) 'Delivering clinical preventive services is a systems problem', *Annals of Behavioral Medicine*, 19(3), pp. 271–278. doi: 10.1007/BF02892291.	Not a systems approach, no comparator
62	Spijker A, Verhey F, Graff M, et al. Systematic care for caregivers of people with dementia in the ambulatory mental health service: Designing a multicentre, cluster, randomized, controlled trial. *BMC Geriatr*. 2009;9(1):1-14. doi:10.1186/1471-2318-9-21	Protocol
63	Vats A, Goin KH, Villarreal MC, Yilmaz T, Fortenberry JD, Keskinocak P. The impact of a lean rounding process in a pediatric intensive care unit. *Crit Care Med*. 2012;40(2):608-617. doi:10.1097/CCM.0b013e318232e2fc	Not a systems approach
64	Vergales BD, Dwyer EJ, Wilson SM, et al. NASCAR pit-stop model improves delivery room and admission efficiency and outcomes for infants <27 weeks' gestation. *Resuscitation*. 2015;92:7-13. doi:10.1016/j.resuscitation.2015.03.022	Systematic but not a systems approach
65	Warner CJ, Walsh DB, Horvath AJ, et al. Lean principles optimize on-time vascular surgery operating room starts and decrease resident work hours. *J Vasc Surg*. 2013;58(5):1417-1422. doi:10.1016/j.jvs.2013.05.007	Not framed in a systems context, a narrowed application of lean
66	Carr, H. *et al.* (2019) 'A Systems-wide approach to prevention of in-hospital newborn falls', *MCN, The American Journal of Maternal/Child Nursing*, 44(2), pp. 100–107.	Not a systems approach
67	Carayon, P. *et al.* (2017) 'Medication Safety in Two Intensive Care Units of a Community Teaching Hospital After Electronic Health Record Implementation: Sociotechnical and Human Factors Engineering Considerations', *Journal of Patient Safety*, 00(00), pp. 1–11. doi: 10.1097/PTS.0000000000000358.	Not a systems approach
68	Scuffham, P. A. *et al.* (2017) 'Evaluation of the Gold Coast Integrated Care for patients with chronic disease or high risk of hospitalisation through a non-randomised controlled clinical trial: A pilot study protocol', *BMJ Open*, 7(6). doi: 10.1136/bmjopen-2017-016776.	Protocol
69	Cumbler, E. *et al.* (2012) 'Improving stroke alert response time: Applying quality improvement methodology to the inpatient neurologic emergency', *Journal of Hospital Medicine*, 7(2), pp. 137–141. doi: 10.1002/jhm.984.	Not set in a systems context
70	Firman, N. and Radrekusa, J. (2016) 'A systems approach to improving cancer screening outcomes through quality improvement strategies', *Journal of Gastroenterology and Hepatology (Australia)*. Netherlands: Blackwell Publishing, 31, pp. 54–55. Available at: http://ovidsp.ovid.com/ovidweb.cgi?T=JS&PAGE=reference&D=emed18&NEWS=N&AN=612984556.	Protocol
71	Lipshutz, A. *et al.* (2015) 'The effect of a comprehensive unit-based safety program on systems thinking in adult ICU providers', 43(12), p. 2015.	Abstract
72	Chrysanthaki, T., Hendy, J. and Barlow (2013) 'Stimulating whole system redesign: Lessons from an organisational analysis of the whole system demonstrator programme', *Journal of health services research & policy*, 18, pp. 47–55.	No quantitative data to analyse